Birthday Paradox and Axioms of Probability

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MATH 241

Lecture #4

Reminders!

- **Homework #1** is due on Friday, 9/11 at 11:59pm
  - Probability problems plus an R problem
- **Checkpoint #1** at the end of class!
  - 1-2 questions, on your own, use of written/printed notes only
- Office hours: Monday from 1-2:30pm in Rocky 106A

Learning Objectives

- Explore the birthday paradox and how we can solve it
- Conceptualize the axiomatic definition of probability and how its consequences allow us to solve probabilities

The Birthday Paradox

Question: At least how many people need to be in a room such that there is a 50% chance that any two people in the room share the same birthday?

Let's watch a video from Vsauce about the paradox:

- Link

The Birthday Problem, A Summary

$$
\begin{aligned}
P(\text{No people sharing birthday}) &= \frac{365}{365} \times \frac{364}{365} \times \cdots \times \frac{365-k+1}{365} \\
&= \frac{365 \times 364 \times \ldots (365-k+1)}{365^k}
\end{aligned}
$$

$$
\implies P(\geq 2 \text{ people sharing birthday}) = 1 - \frac{365 \times 364 \times \ldots (365-k+1)}{365^k}
$$

The Birthday Problem, A Summary

When $k = 23$:

$$
\begin{aligned}
P(\geq 2 \text{ people sharing birthday}) &= 1 - \frac{365 \times 364 \times \ldots (365 - 23 + 1)}{365^{23}} \\
&= 1 - 0.4927 \\
&= 0.5073
\end{aligned}
$$

The Birthday Problem, A Summary

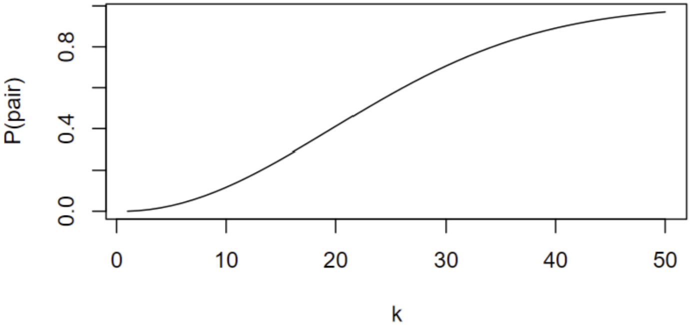

Figure: $P(\geq 2$ people sharing birthday) for $k \in \{1, 2, \ldots, 50\}$

k	10	20	30	40	50
Prob.	0.1169	0.4114	0.7063	0.8912	0.9704

Activity: The Birth **Hour** Problem

Pair up with someone around you and consider the following: you are looking to find the probability that any two people share the same hour of the day they were born in. Assume there is a uniform distribution of babies born throughout a day, meaning that there is an equal chance a baby is born in any given hour.

Do the following:

1. Identify what would change from the birthday problem to this problem.
2. For 4, 8, and 12 people in a classroom: find $P(\geq 2$ people sharing birthhour$)$.
3. Similar to the birthday problem, find the number of people needed such that $P(\geq 2$ people sharing birthhour$) \geq 0.5$.

Example: The Binomial Theorem

Let $n \in \mathbb{N}$ and let x, y be variables. Then, the binomial theorem states that:

$$(x+y)^n = \sum_{i=1}^{n} \binom{n}{k} x^k y^{n-k}$$

Example: The Binomial Theorem

For $n = 2$:

$$
\begin{aligned}
(x+y)^2 &= \sum_{k=0}^{2} \binom{2}{k} x^k y^{2-k} \\
&= \binom{2}{0} x^0 y^2 + \binom{2}{1} x^1 y^1 + \binom{2}{2} x^2 y^0 \\
&= y^2 + 2xy + x^2 \\
&= x^2 + 2xy + y^2
\end{aligned}
$$

Later: We will see that this will motivate the use of finding binomial probabilities, when we deal with repeated, independent binary trials.

Naive Definition of Probability: A Look Again

Suppose all outcomes of an experiment are equally-likely, then the probability of an event A is given by:

$$P(A) = \frac{\text{number of outcomes in A}}{\text{number of outcomes in S}}$$

We need to generalize this definition for an infinite number of outcomes, while also formalizing it.

General Definition of Probability

We have some probability space, which consists of...

- A sample space S (set of all outcomes)
- A probability function P (mapping an event $A \subseteq S$ to $P(A)$)
  - $0 \leq P(A) \leq 1$

What are the consequences here?

General Definition of Probability

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What are the consequences here?

- $P(\emptyset) = 0$, $P(S) = 1$
- If $A_1, A_2, \ldots$ are disjoint events, then $P(\bigcup_{j=1}^{\infty} A_j) = \sum_{j=1}^{\infty} P(A_j)$
  - Disjoint $=$ mutually exclusive
  - $A_i \cap A_j = \emptyset$ for $i \neq j$

Frequentist vs. Bayesian

- Frequentist: probability of a coin lands head is 50% means if we toss the coin many times, then half the time it would land heads
- Bayesian: probability as a degree of belief

Example

Let $S = \{1, 2\}$ where...

- $P(\emptyset) = 0$,
- $P(\{1\}) = 0.5$,
- $P(\{2\}) = 0.5$, and
- $P(\{1, 2\}) = 1$.

Verify that this probability function satisfies all the axioms of probability.

Properties of Probability

For any events A and B, we have the following properties:

Complement:

- $P(A^C) = 1 - P(A)$

Subset:

- If $A \subseteq B$, then $P(A) \leq P(B)$.

Union:

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Properties of Probability: Proof #1

$$P(A^C) = 1 - P(A)$$

Proof.

Since A and A^C are disjoint and their union is S, the second axiom then gives the following:

$$P(S) = P(A \cup A^C) = P(A) + P(A^C)$$

Since $P(S) = 1$, then $P(A) + P(A^C) = 1$. This implies that $P(A^C) = 1 - P(A)$. $\square$

Properties of Probability: Proof #2

If $A \subseteq B$, then $P(A) \leq P(B)$.

Proof.

Since $A \subseteq B$, then we can write B as a union of A and $B \cap A^C$. Since A and $B \cap A^C$ are disjoint, we apply the second axiom:

$$P(B) = P(A \cup (B \cap A^C)) = P(A) + P(B \cap A^C)$$

$P(B \cap A^C) \geq 0$, which means that $P(B) \geq P(A)$. $\square$

Properties of Probability: Proof #3

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Proof.

Write $P(A \cup B)$ as the following:

$$\begin{aligned} P(A \cup B) &= P(A) \cup (B \cap A^C)) \\ &= P(A) + P(B \cap A^C) \qquad \text{by axiom \#2} \end{aligned}$$

$P(B \cap A^C) = P(B) - P(A \cap B)$ by definition. Hence,

$$\begin{aligned} P(B) - P(A \cap B) &= P(A \cup B) - P(A) \\ \implies P(A \cup B) &= P(A) + P(B) - P(A \cap B) \end{aligned}$$

$\square$

Suppose the probability function for this experiment has:

$$P(\{a\}) = \frac{1}{2}$$
$$P(\{b\}) = \frac{1}{4}$$
$$P(\{c\}) = \frac{1}{8}$$
$$P(\{d\}) = \frac{1}{8}$$

What is the probability that neither A nor B occurs?

Inclusion-Exclusion Principle

Its use: Finding probability of finding the probability of a union of events that are not disjoint.

$$\text{For } n = 2: P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\text{For } n = 3: P(A \cup B \cup C) = P(A) + P(B) + P(C)$$
$$- P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

$$\text{In general}: P\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{i=1}^{n} P(A_i) - \sum i < j P(A_i \cap A_j)$$
$$+ \sum_{i=1}^{n} P(A_i \cap A_j \cap A_k) - \cdots + (-1)^{n+1} P(A_1 \cap \cdots \cap A_n)$$

Checkpoint!

- You may use handwritten or typed notes on paper for this checkpoint. No electronics are permitted.
- Read and follow all directions; points will be deducted if not all work is shown.
- You have until the rest of class to finish the questions.

See You Tuesday!

- Homework #1 is due on Friday, 9/11 at 11:59pm
  - Probability problems
- Read sections 2.1 to 2.3 from *Introduction to Probability*
- Office hours: Monday from 1-2:30pm in Rocky 106A